EDU GENIE

Google Gemini Powered Learning Assistant

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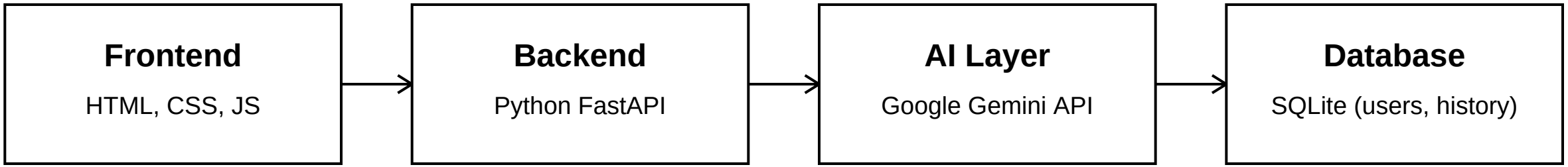
Project Description

- Students often find it hard to get quick and simple answers for their doubts.
- **Edu Genie** is a learning assistant website powered by Google Gemini AI.
- It explains topics in simple words, summarizes notes, creates quizzes and makes study plans.
- Students can type a question or upload an image of a problem and get help at once.
- It makes learning easy, fast and personal for every student.

Scenarios

- **Scenario 1: Doubt Solver.** The student asks a question. Gemini gives a step-by-step answer in easy language.
- **Scenario 2: Notes Summarizer.** The student pastes long notes. The AI gives short points for quick revision.
- **Scenario 3: Quiz and Study Plan.** The student enters a topic and exam date. The AI creates practice questions and a daily study plan.

Architecture Overview



- The user sends a question from the web page to the FastAPI server.
- The server sends a prompt to Gemini and gets the answer back.
- The answer is shown on screen and saved in the history.

Pre-requisites and Workflow

Pre-requisites

- Python basics
- Gemini API key (Google AI Studio)
- FastAPI framework
- VS Code and GitHub
- Render account for deployment

Project Workflow

1. Set up Gemini API
2. Build core features
3. Create FastAPI backend
4. Design the user interface
5. Test and improve
6. Deploy on Render

Milestones

Milestone 1	**Gemini AI Setup** Get API key, test text and image prompts.
Milestone 2	**Core Functions** Doubt solver, notes summarizer, quiz maker, study plan.
Milestone 3	**Backend** FastAPI routes, login, register and history.
Milestone 4	**User Interface** Home, login, dashboard, tools and result pages.
Milestone 5	**Testing** Check answers, fix errors and improve prompts.

Conclusion

- Edu Genie uses Google Gemini to make learning simple and quick.
- It helps students solve doubts, revise notes and practise with quizzes anytime.
- The FastAPI backend and easy web pages make it fast and simple to use.
- In future, we can add voice support, more languages and progress reports.
- **Thank you!**